

Caden Sun

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SKILLS

Languages: Python, JavaScript/TypeScript, Java, C/C++, SQL, HTML/CSS

Frameworks: PyTorch, Pandas, NumPy, Matplotlib, Scikit-learn, XGBoost, Flask, React, Node.js, LangGraph

Tools & Platforms: AWS (S3, EC2, DynamoDB), Docker, CI/CD, Jupyter Notebook, Linux, Git, Vercel

EDUCATION

University of Waterloo

Sep 2025 – Apr 2030

Bachelor of Mathematics

Relevant Coursework: Data Structures & Algorithms, Designing Functional Programs, Linear Algebra, Probability & Statistics

EXPERIENCE

AI Intern | *ProCogia*

May 2026 – Present

- Built an **AI calling agent POC** from scratch using a **LangGraph** orchestrated LLM, reducing latency by around **500 ms** (a **3× improvement** over the Amazon Connect version).
- Built a **RAG** pipeline on **AWS** connecting a document store in **S3** to an LLM context window.
- Exhaustively tested** the full call agent pipeline and **API endpoints** by writing live-monitoring **Python** scripts, catching **12 major bugs** in the process.
- Conducted **root-cause analyses** on pipeline failures by tracing **HTTP request/response chains** and inspecting status codes and payloads, **isolating all major bugs** across the backend, **API endpoints**, and auth layer.

ML Engineer | *Wat Street (University of Waterloo's Quantitative Finance Team)*

Jan 2026 – Present

- Built the **data pipeline** converting raw S&P 500 **OHLCV** data into model-ready inputs using **Pandas** and **NumPy** by engineering **10 features** for each stock over a **30-day window**.
- Implemented **missing-data handling** cases with shape and **NaN validation**, producing clean **tensor inputs** for up to **500 stocks**.
- Connected a **Graph Attention Network** to an **LSTM** in **PyTorch** by writing the module that converts the GAT's outputs for each stock into the tensor format the LSTM requires, **enabling the volatility forecasts** for 500 stocks.

Software Developer | *Waterloo Aerial Robotics Group (WARG)*

Oct 2025 – Apr 2026

- Implemented client-side mission export functionality in **JavaScript** and **React** for **IMACS-3.0**, enabling users to save mission waypoint data directly from the ground control station UI.
- Generated **QGC-compatible .waypoints** files by formatting mission data into QGC WPL 110 format, adding coordinate conversion logic and empty-state validation to prevent invalid exports.

PROJECTS

Flight Price Predictor | *Python, SQLite, XGBoost, Pandas, NumPy, Scikit-learn* | 🌐

- Currently building an end-to-end flight price prediction system covering **200+ routes** across **55+ airports**.
- Built an **ingestion pipeline** pulling **~4,000 flight offers/day** into an **SQLite** database with **automated cloud backup** during a planned **four-month** collection period, **reducing failure rate to 0%** across multiple slow-API days.
- Decided on **18 features** after **Jupyter Notebook** graphs revealed the most strongly correlated features (route distance was found to be the highest correlated feature for most flights) during **careful data exploration**.
- Training an **XGBoost gradient-boosted model** on chosen features and a **70/15/15 training, validation, and testing** split, currently at **14%** percentage error (**targeting sub-10%**).
- Maintained **dataset quality** by writing automated **SQL deduplication** and **integrity-check scripts** running daily after ingestion, including a **9-column exact-key dedup guard**, **NULL** checks on important feature fields, and **volume-drop detection** against a 3-day rolling baseline; **0 detected anomalies** across **180K+ rows** to date.

AI Chess Bot | *Python, PyTorch, Flask, React/TypeScript, Docker* | 🌐 | 🏠

- Deployed a **full-stack chess AI** on **Vercel** (containerized with **Docker** for an alternative **Render** deployment), serving a **PyTorch** neural network (**768 → 256 → 128 → 1, ReLU**) via **Flask REST API** and **React/TypeScript** frontend with **sub-second** response times and public accessibility.
- Improved **model generalization** (**30% reduction** in validation loss) across **4 training iterations** from a baseline evaluator to a tactical model using Stockfish-labeled data, **Adam optimizer**, **batch normalization**, and hand-crafted signals: material balance, castling rights, and center control.